

Olist E-Commerce Analytics Project

Business Analysis, Power BI Dashboard & Insights

Executive Summary

This project analyzes the Olist Brazilian e-commerce dataset to evaluate sales performance, customer behavior, delivery efficiency, product performance, and seller performance. Python was used for data preparation and exploratory analysis, while Power BI and DAX were used to build the final interactive five-page dashboard.

KPI Summary

KPI	Result
Total Revenue	~  13.59M
Total Orders	~99.44K
Unique Customers	~96.10K
Total Sellers	3,095
Total Products	32,951
Average Order Value	~  136.68
Delivered Orders	~96.48K
Late Orders	~6.54K
Late Delivery Rate	~6.7%
Repeat Customers	~2,997 (~3.1%)

1. Project Objective

Transform raw Olist e-commerce data into actionable business intelligence across executive performance, sales, customers, delivery, products, and sellers.

2. Business Questions

Sales: revenue trends, state and category drivers, and volume versus value. Customers: repeat purchasing, customer value, and satisfaction. Delivery: late orders, high-delay states, delivery time, and satisfaction. Sellers/products: top contributors and multidimensional performance.

3. Data Preparation and Methodology

Workflow: Raw Olist Dataset → Data Cleaning → EDA → Integrated Business Analysis → Power BI Data Model → DAX Measures → Interactive Dashboard. Python included Pandas, NumPy, Matplotlib, and Seaborn. Power BI was used for modeling, DAX, KPIs, filtering, and dashboard design.

4. Data Model

Core dimensions include DimCustomer, DimOrder, DimProduct, DimSeller, and DimDate, supported by sales, payment, delivery, and customer-satisfaction analysis tables. Relationships were checked for grain, uniqueness, cardinality, active status, and filter direction.

5. Page 1 — Executive Dashboard

Provides high-level KPIs, revenue and order trends, revenue by state, delivery status, category performance, satisfaction, and executive insights. SP is the dominant revenue market (~\$5.20M); late delivery is ~6.7%; repeat customers are ~3.1%.

6. Page 2 — Sales & Revenue

Analyzes monthly revenue, order volume, top categories, revenue by state, and revenue versus order volume. SP is the largest revenue market, while market volume does not necessarily imply the highest revenue per customer.

7. Page 3 — Customer Insights

Approximately 93,099 customers were one-time purchasers and 2,997 were repeat purchasers, producing a repeat-customer rate of ~3.1%. Average orders per customer were ~1.00 for one-time customers versus ~2.12 for repeat customers. Retention is a major growth opportunity.

8. Page 4 — Delivery Performance

Approximately 6.7% of delivered orders were late. AL had the highest late-delivery rate (~20.58%) and average delivery time (~23.99 days) in the project analysis. Other elevated-delay states included MA, SE, PI, CE, BA, and PB.

9. Page 5 — Seller & Product Performance

Identifies top sellers, products, category contribution, seller revenue versus order volume, and multidimensional seller performance. Revenue, volume, AOV, delivery, and satisfaction should be evaluated together.

10. Key Business Insights

Revenue is concentrated in SP and leading categories; repeat purchasing is low; delivery performance varies substantially by state; smaller states can have higher revenue per customer; and seller performance is multidimensional.

11. Business Recommendations

Improve retention through personalized offers and re-engagement. Prioritize high-delay states. Evaluate high-value markets for targeted expansion. Strengthen category strategy. Monitor sellers across revenue, orders, AOV, delivery time, and review score.

12. Dashboard Pages

1. Executive Dashboard 2. Sales & Revenue 3. Customer Insights 4. Delivery Performance 5. Seller & Product Performance

13. Conclusion

The project demonstrates an end-to-end analytics workflow from Python data preparation and exploratory analysis to Power BI dimensional modeling, DAX, and interactive reporting. Major opportunities are retention, delivery improvement, geographic expansion, category optimization, and seller performance management.

Currency note: The source Olist dataset is denominated in Brazilian reais (BRL). If publishing externally, label currency consistently or document any conversion used.